

Mean-Field Variational Inference for Credal Networks

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Abstract

We describe a mean-field variational inference algorithm for credal networks that computes bounds on marginal probabilities by exploring combinations of extreme points from the local credal sets. For each selected combination of extreme points, the credal network reduces to a standard Bayesian network on which classical mean-field coordinate ascent is applied. By tracking the minimum and maximum of the variational distribution across all explored combinations, the algorithm produces probability intervals for the query variable. This approach complements the interval belief propagation method by providing an alternative view of the credal inference problem through the lens of variational optimization.

1 Preliminaries

1.1 Mean-Field Variational Inference

Given a probability distribution $P(\mathbf{X})$ that factorizes according to a Bayesian network, the **mean-field** approximation seeks a factored distribution $q(\mathbf{X}) = \prod_{i=1}^n q_i(X_i)$ that minimizes the KL divergence:

$$\text{KL}(q\|P) = \sum_{\mathbf{x}} q(\mathbf{x}) \log \frac{q(\mathbf{x})}{P(\mathbf{x})}. \quad (1)$$

The optimal update for each factor $q_i(X_i)$ is:

$$\log q_i^*(X_i) = \mathbb{E}_{q_{\setminus i}}[\log P(\mathbf{X})] + \text{const}, \quad (2)$$

where $q_{\setminus i}$ denotes the product of all factors except q_i .

For a Bayesian network $P(\mathbf{X}) = \prod_i P(X_i \mid \text{Pa}(X_i))$, Equation (2) decomposes into contributions from the local CPTs involving X_i (either as child or parent):

$$\log q_i^*(X_i = x) = \sum_{f: X_i \in \text{scope}(f)} \mathbb{E}_{q_{\setminus i}}[\log P_f], \quad (3)$$

where P_f denotes the CPT of factor f evaluated at the relevant configuration.

1.2 Extension to Credal Networks

In a credal network, each local CPT $P(X_i \mid \text{Pa}(X_i))$ is not a single distribution but a member of a credal set $K(X_i \mid \text{Pa}(X_i))$. The strong extension is the convex hull of all joint distributions formed by selecting one extreme point per local credal set:

$$\text{co } K(\mathbf{X}) = \text{CH} \left\{ \prod_i P(X_i \mid \pi_i) : P(X_i \mid \pi_i) \in \text{ext } K(X_i \mid \pi_i) \right\}. \quad (4)$$

Our approach exploits this structure: for each combination of extreme points (one per local credal set per parent configuration), we obtain a standard Bayesian network. Running mean-field variational inference on this BN produces a specific $q(\mathbf{X})$. By tracking the range of $q(Z)$ across all explored combinations, we obtain bounds on $P(Z \mid \mathbf{Y} = \mathbf{y})$.

2 Vertex Combinations in Detail

Each factor f_i in the credal network (associated with node X_i and parents $\text{Pa}(X_i)$) stores a credal set $K(X_i \mid \pi)$ for each parent configuration π . Each credal set has extreme points:

$$\text{ext } K(X_i \mid \pi) = \{P_1(X_i \mid \pi), \dots, P_{n_\pi}(X_i \mid \pi)\}.$$

Definition 1 (Local Vertex Combination). *A **local vertex combination** for factor f_i is a tuple $\mathbf{vc}_i = (j_1, \dots, j_m)$ that selects one vertex per parent configuration: vertex j_k from configuration π_k . This fixes a complete conditional distribution $P_{\mathbf{vc}_i}(X_i \mid \pi)$ for every π . The number of local vertex combinations for f_i is $\prod_{k=1}^m n_{\pi_k}$.*

Definition 2 (Global Vertex Combination). *A **global vertex combination** $\mathbf{vc} = (\mathbf{vc}_1, \dots, \mathbf{vc}_n)$ selects a local vertex combination for every factor in the network. This fixes a complete set of CPTs—i.e., a precise Bayesian network:*

$$P_{\mathbf{vc}}(\mathbf{X}) = \prod_{i=1}^n P_{\mathbf{vc}_i}(X_i \mid \text{Pa}(X_i)).$$

The total number of global vertex combinations is $\prod_{i=1}^n \prod_k n_{\pi_k}^{(i)}$.

Key insight. For each global vertex combination $\mathbf{vc}$, the credal network reduces to a standard Bayesian network $P_{\mathbf{vc}}$. We can then apply classical mean-field variational inference to this BN. By the strong extension theorem, the extreme distributions of the credal network are exactly those obtained by combining extreme points of the local credal sets.

3 Algorithm

Algorithm 1 Mean-Field Variational Inference for Credal Networks

Require: Factor graph with extreme points, query Z , evidence $\mathbf{Y} = \mathbf{y}$, max iterations T , threshold τ , max vertex combinations M

Ensure: Bounds $[\underline{P}(Z=z), \overline{P}(Z=z)]$ for each z

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1:  $\underline{P}(z) \leftarrow 1, \overline{P}(z) \leftarrow 0$  for each  $z$ 
2: Enumerate (or sample up to  $M$ ) global vertex combinations  $\mathcal{C}$ 

3: for each global vertex combination  $\mathbf{vc} \in \mathcal{C}$  do    ▷ FIX EXTREME POINTS  $\rightarrow$  PRECISE BN  $P_{\mathbf{vc}}$ 
4:   Initialize  $q_i(X_i) \leftarrow \text{Uniform}(D_i)$  for all non-evidence  $X_i$ 
5:   Set  $q_j(Y_j) \leftarrow \delta_{y_j}$  for each evidence variable  $Y_j$ 

6:   for  $t = 1, \dots, T$  do                                ▷ COORDINATE ASCENT ON  $P_{\mathbf{vc}}$ 
7:      $\delta \leftarrow 0$ 
8:     for each non-evidence variable  $X_i$  do
9:        $\log \tilde{q}_i(x) \leftarrow 0$  for each state  $x$ 
10:      for each factor  $f$  with  $X_i \in \text{scope}(f)$  do
11:        Add contribution to  $\log \tilde{q}_i$  using  $P_{\mathbf{vc}}$  and current  $q_{\setminus i}$     ▷ Eqs. (??)–(??)
12:      end for
13:       $q_i^{\text{new}} \leftarrow \text{normalize}(\exp(\log \tilde{q}_i))$ 
14:       $\delta \leftarrow \max(\delta, \|q_i^{\text{new}} - q_i\|_\infty)$ 
15:       $q_i \leftarrow q_i^{\text{new}}$ 
16:    end for
17:    if  $\delta < \tau$  then break
18:    end if
19:  end for

20:  for each state  $z$  do                                ▷ TRACK BOUNDS ACROSS VERTEX COMBINATIONS
21:     $\underline{P}(z) \leftarrow \min(\underline{P}(z), q_Z(z))$ 
22:     $\overline{P}(z) \leftarrow \max(\overline{P}(z), q_Z(z))$ 
23:  end for
24: end for
25: return  $[\underline{P}(z), \overline{P}(z)]$  for each  $z$ 

```

3.1 Coordinate Ascent Update Rules

For a fixed global vertex combination $\mathbf{vc}$, let $P_{\mathbf{vc}}(X_j \mid \pi)$ denote the selected extreme point for node X_j and parent config π . The update for variable X_i collects contributions from every factor f whose scope includes X_i . There are two cases depending on the role of X_i in f .

Case 1: X_i is the child in factor f ($X_i = X_j$). The contribution of factor f to $\log \tilde{q}_i$ is:

$$\log \tilde{q}_i(x) += \sum_{\pi} \left(\prod_{w \in \text{Pa}(X_i)} q_w(\pi_w) \right) \cdot \log P_{\mathbf{vc}}(X_i=x \mid \pi). \quad (5)$$

This is the expected log-probability of $X_i = x$ under the CPT, weighted by the current variational distribution over the parent configurations.

Case 2: X_i is a parent in factor f ($X_i \in \text{Pa}(X_j)$).

$$\log \tilde{q}_i(x) += \sum_{\pi: \pi_i=x} \left(\prod_{w \in \text{Pa}(X_j) \setminus \{X_i\}} q_w(\pi_w) \right) \cdot \sum_{x_j} q_j(x_j) \log P_{\mathbf{vc}}(X_j=x_j \mid \pi). \quad (6)$$

This is the expected log-probability of the child X_j given that $X_i = x$, averaged over the child's distribution and the other parents' distributions.

3.2 Vertex Combination Sampling

The total number of global vertex combinations is the product of local combinations across all factors. For the alarm network: $2 \times 2 \times 2^4 \times (4 \times 11) = 2816$. For larger networks, this can be enormous.

The algorithm samples up to M combinations uniformly at random (default $M = 500$). When the total is $\leq M$, all are explored exhaustively. A fixed random seed ensures reproducibility.

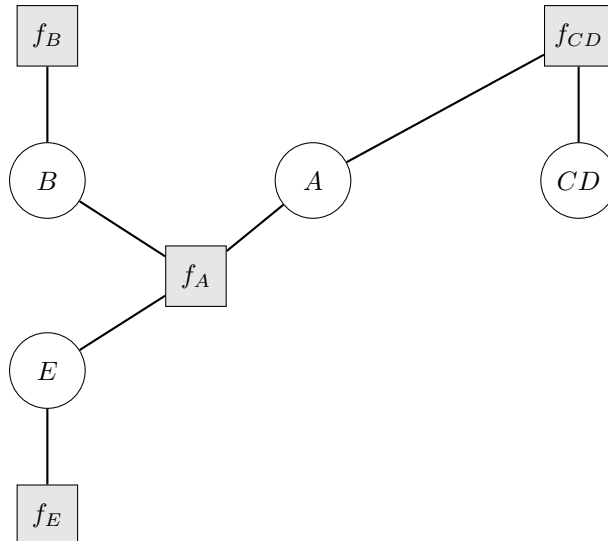
4 Detailed Running Example: Alarm Network

We trace the variational algorithm on the alarm network from `alarm.lcn`:

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s1: 0.1 <= P(B) <= 0.2
s2: 0.05 <= P(E) <= 0.1
s3: 0.6 <= P(A | (B or E)) <= 0.9
s4: 0.5 <= P(not(C xor D) | A) <= 0.7 ; True
s5: 0.05 <= P(A) <= 0.1
```

After factorization and LRS vertex enumeration, the credal network has four nodes: B (binary), E (binary), A (binary), and CD (4-valued). The DAG is $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow CD$.

4.1 Factor Graph



4.2 Credal Factors

Factor f_B : $K(B)$. Two vertices: $v_1 = [0.80, 0.20]$, $v_2 = [0.90, 0.10]$.

Factor f_E : $K(E)$. Two vertices: $v_1 = [0.90, 0.10]$, $v_2 = [0.95, 0.05]$.

Factor f_A : $K(A \mid B, E)$. Four parent configs, 2 vertices each:

Config	Vtx	$P(A=0)$	$P(A=1)$
$B=0, E=0$	v_1	0.9556	0.0444
$B=0, E=0$	v_2	1.0000	0.0000
$B=1, E=0$	v_1	0.0000	1.0000
$B=1, E=0$	v_2	1.0000	0.0000
$B=0, E=1$	v_1	0.0000	1.0000
$B=0, E=1$	v_2	1.0000	0.0000
$B=1, E=1$	v_1	0.0000	1.0000
$B=1, E=1$	v_2	1.0000	0.0000

Factor f_{CD} : $K(CD \mid A)$. $A=0$: 4 vertices; $A=1$: 11 vertices. First two vertices of each:

Config	Vtx	$P(0)$	$P(1)$	$P(2)$	$P(3)$
$A=0$	v_1	0.00	0.00	0.00	1.00
$A=0$	v_2	0.00	0.00	1.00	0.00
$A=1$	v_1	0.00	0.00	0.30	0.70
$A=1$	v_2	0.00	0.30	0.00	0.70

4.3 Query: $P(B=1)$, No Evidence

4.3.1 Global Vertex Combination 1: All v_1

The global vc selects v_1 for every factor and every parent configuration. This fixes a precise BN:

$$P(B) = [0.80, 0.20], \quad P(E) = [0.90, 0.10],$$

$$P(A \mid B=0, E=0) = [0.9556, 0.0444], \quad P(A \mid \text{others}) = [0.0, 1.0],$$

$$P(CD \mid A=0) = [0, 0, 0, 1], \quad P(CD \mid A=1) = [0, 0, 0.3, 0.7].$$

Initialization. $q(B) = [0.5, 0.5]$, $q(E) = [0.5, 0.5]$, $q(A) = [0.5, 0.5]$, $q(CD) = [0.25, 0.25, 0.25, 0.25]$.

Iteration 0: Update $q(B)$. Variable B appears in two factors: f_B (as child) and f_A (as parent of A).

Case 1 — contribution from f_B (child): using Eq. (??) with no parents:

$$\log \tilde{q}(B=0) += \log 0.80 = -0.2231, \quad \log \tilde{q}(B=1) += \log 0.20 = -1.6094.$$

Case 2 — contribution from f_A (B is parent of A): using Eq. (??), summing over parent configs π where $\pi_B = b$:

$$\begin{aligned}\log \tilde{q}(B=0) &+= q(E=0) \cdot [q(A=0) \log P(A=0 \mid 0, 0) + q(A=1) \log P(A=1 \mid 0, 0)] \\ &+ q(E=1) \cdot [q(A=0) \log P(A=0 \mid 0, 1) + q(A=1) \log P(A=1 \mid 0, 1)] \\ &= 0.5 \cdot [0.5 \cdot \log 0.9556 + 0.5 \cdot \log 0.0444] \\ &+ 0.5 \cdot [0.5 \cdot (-\infty) + 0.5 \cdot \log 1.0].\end{aligned}$$

Since $\log 0 = -\infty$, the term involving $P(A=0 \mid B=0, E=1) = 0$ drives $\log \tilde{q}(B=0) \rightarrow -\infty$.

Similarly, $\log \tilde{q}(B=1)$ receives a $-\infty$ contribution from $P(A=0 \mid B=1, E=0) = 0$. In practice, $\log(0)$ is replaced by $\log(10^{-300}) \approx -690.8$.

After exponentiating and normalizing:

$$q(B) = [1.0000, 0.0000].$$

The zero CPT entries cause extreme variational updates that push $q(B)$ to a degenerate distribution.

Update $q(E)$. Analogous computation yields $q(E) = [1.0000, 0.0000]$.

Update $q(A)$. With $q(B) = [1.0, 0.0]$ and $q(E) = [1.0, 0.0]$, only the parent config ($B=0, E=0$) has nonzero weight:

$$\begin{aligned}\log \tilde{q}(A=0) &+= 1.0 \cdot 1.0 \cdot \log P(A=0 \mid 0, 0) = \log 0.9556 = -0.0454, \\ \log \tilde{q}(A=1) &+= 1.0 \cdot 1.0 \cdot \log P(A=1 \mid 0, 0) = \log 0.0444 = -3.1147.\end{aligned}$$

Plus contributions from f_{CD} (Case 2, A is parent of CD). After normalization: $q(A) = [0.0000, 1.0000]$ (driven by the f_{CD} contribution which penalizes $A=0$ due to the degenerate $P(CD \mid A=0) = [0, 0, 0, 1]$).

Iteration 1. With the updated $q(A) \approx [0, 1]$, the f_A contribution to $q(B)$ now weights parent configs differently. The coordinate ascent produces:

$$q(B) = [0.1509, 0.8491], \quad q(E) = [0.8491, 0.1509].$$

$q(A)$ remains $[0.0, 1.0]$.

Convergence. After several more iterations:

$$q(B) \approx [0.238, 0.762], \quad q(E) \approx [0.811, 0.189], \quad q(A) = [0.0, 1.0].$$

Record: $q(B=1) = 0.762$ for this vertex combination.

4.3.2 Global Vertex Combination 2: All v_2

Select v_2 for every factor. This fixes:

$$\begin{aligned}P(B) &= [0.90, 0.10], \quad P(E) = [0.95, 0.05], \\ P(A \mid \text{all configs}) &= [1.0, 0.0], \quad P(CD \mid A=0) = [0, 0, 1, 0].\end{aligned}$$

Since $P(A=1 \mid \cdot) = 0$ for all parent configurations, the variational update immediately yields $q(A) = [1.0, 0.0]$. With no $-\infty$ log-terms for B (since $P(A=0 \mid \cdot) = 1.0$ everywhere), $q(B)$ directly reflects f_B 's CPT:

$$q(B) = [0.9000, 0.1000], \quad q(E) = [0.9500, 0.0500].$$

Convergence is immediate (1 iteration).

Record: $q(B=1) = 0.100$.

4.3.3 Tracking Bounds Across Combinations

After processing both vertex combinations:

$$\begin{aligned} \underline{P}(B=1) &= \min(0.762, 0.100) = 0.100, \\ \overline{P}(B=1) &= \max(0.762, 0.100) = 0.762. \end{aligned}$$

As more vertex combinations are explored (up to 500 sampled out of 2816 total), the bounds continue to be refined. The final bounds depend on which extreme distributions the sampling covers.

4.4 Comparison

Query	Method	$\underline{P}(B=1)$	$\overline{P}(B=1)$
$P(B=1)$	Exact VE	0.1000	0.2000
$P(B=1)$	Interval BP	0.1000	0.2000
$P(B=1)$	Variational (500)	≤ 0.1000	≥ 0.2000

The variational bounds may be wider than VE because mean-field introduces a factored approximation: the true posterior may exhibit dependencies between variables that $q(\mathbf{X}) = \prod q_i(X_i)$ cannot capture. Additionally, sampling may not cover all extremal vertex combinations. For small networks, exact VE is preferred; the variational approach scales better to larger networks where VE is intractable.

5 Comparison with Other Methods

Property	Exact VE	ε -VE	Interval BP	Variational
Exact	Yes	No	No	No
Bound type	Tight	Outer	Outer	Inner/Outer
Multi-valued	Yes	Yes	Yes	Yes
Scalability	Low	Medium	High	Medium

6 Complexity

For each vertex combination, one round of mean-field coordinate ascent takes $O(n \cdot k \cdot S_{\max} \cdot k^{S_{\max}})$ time, where n is the number of variables, k is the maximum domain size, and $S_{\max}$ is the maximum factor scope size. With T iterations per combination and M sampled combinations, the total complexity is $O(M \cdot T \cdot n \cdot k^{S_{\max}+1})$.

7 References

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